

SAFE WORK METHOD STATEMENT (SWMS)

In order of priority:

1. **ELIMINATION** – Remove hazard completely.
2. **Substitution** – Replace object /process with less hazardous one.
3. **Isolation** – Isolate the hazard (i.e. separate the hazard from the people).
4. **Engineering** – Modify a process (i.e. redesign a workstation).
5. **Administration** – Use procedures /systems to minimise the risk (i.e. Rotation of employees, implementation of a safe work method, training, etc).
6. **Personal Protective Equipment (PPE)** – use to try to protect the person.

LIKELIHOOD	RISK SCORE CALCULATOR				
Certain to Occur	Mod.	High	High	Extreme	Extreme
Very Likely	Mod.	Mod.	High	Very High	Extreme
Possible	Low	Mod.	High	Very High	Very High
Unlikely	Low	Mod.	Mod.	High	Very High
Rare	Low	Low	Low	Mod.	High
CONSEQUENCE	LOW	MINOR	MODERATE	FATAL	CATASTROPHIC

Residual Risk	Action Required
Low	Ok for now. Continue to monitor.
MODERATE	Determine if remedial action/treatment is required.
HIGH	Immediate decision required by site leadership. Identify and commence remedial action with appropriate priority.
VERY HIGH	ACT NOW. Requires immediate attention by site leadership. Address the risk without delay. Requires urgent attention and clear, timely action plans to reduce the risk before proceeding.
EXTREME	ACT NOW. Requires immediate attention by site leadership. Urgently address the risk. Requires immediate resolution with focused attention until resolved.

Identify task high risk workplace hazards - *Non routine work is work not documented in a Safe Operating Procedure (SOP)* or Work Instruction

Note: If machine isolation is required, workers must be competent in the relevant isolation procedure.

EXTREME & VERY HIGH RISK HAZARDS	Risk	MAIN RISKS / PERMITS REQUIRED
1. Entering a Confined Space.	E	- Heat exhaustion, engulfment, limited access/egress. - Asphyxiation – low O₂ or contaminants. (Confined Space Entry Permit required) - Explosive atmosphere, combustible dusts (Confined Space Entry/Hot Work Permits required)
2. Work in an area which may have a contaminated or flammable atmosphere.	E	- Fire / explosion from ignition source. (Hot Work Permit required) - Asphyxiation or poisoning from contaminated atmosphere. (Confined Space Permit)
3. Work with potential for a fall from 2m or work on a roof	E	Falling worker/object ≥2m requires SWMS.
4. Hot work required. (Oxy cutting, welding, soldering, grinding, etc).	E	- Fire or explosion from ignition source. (Hot Work Permit required) - Injury caused by hot surface, grinding wheel ejected pieces, welders "flash", etc. Fire inside Expanded Polystyrene (EPS) wall panels (Hot Work Permit required).
5. Hand dig >200mm OR ground penetration >200mm OR mechanical earthworks OR enter a trench >1.5 M deep (excavating is Construction work)	E	- Contact utilities – electricity, water, gas, communications. (Excavation Permit required) - Trench Collapse. (Excavation Permit required)
6. Non routine electrical, high voltage or 'live' work or work near electrical installations.	E	- High voltage is >33kV. 'Live' work is when electrical equipment is worked on energised – (Management Approval Required) - Electrical cabinets or MCC identified as Red
7. Non routine work involving mechanical lifting, lowering, pushing or pulling.	E	- High mechanical force damage / injury. - High risk work licence for use of cranes, mobile plant, scaffold, dogging, etc.
8. Using high risk tools with excess noise, kickback, cutting edge, high speed/pressure, ejected material, weight, awkward posture, explosive force.	E	- Injury from use of high risk powered or hand tools. - Electrocution, burn or electric shock. - Manual task injury from weight, forces, overexertion, repetition or poor work heights.
9. Demolish / alter load bearing / integral part of a structure 6m high OR use load shifting machinery on a suspended floor OR use explosives. Includes modify handrail / floors.	E	Being struck or crushed by parts of a structure or exposure to asbestos, lead, hazardous chemicals etc. (Demolition License and/or asbestos removal licence and Safe Work Plan required)
10. Work inside fences / barricades / enclosures where moving / rotating machinery is operating or hazards may exist.	V	- Serious bodily injury caused by contacting moving/unguarded plant or materials. - Machinery starting unexpectedly or electrocution, burn or electric shock.
11. Potential or stored energy which cannot be isolated (electrical, hydraulic, pneumatic, kinetic).	V	- Electrocution or entrapment. - Machinery entanglement / injury.
12. Work on or near a pressurized or enclosed circuit (water, pneumatic, hydraulic, hot oil, etc).	V	- Fracturing the mains or piping from contact – spills / leaks / emissions (environment). - Contacting temperature extremes.
13. Working in artificial temperature extremes or weather extremes. Hot, cold, wet, icy, lightning, high winds.	V	- Electrocution from using power tools / leads in wet conditions or lightning. - Heat stroke / Heat Exhaustion / Fatigue / Frost bite / Hypothermia / Influenza. - Slip / Trip / Fall Hazards.
14. Hazardous manual tasks or excessive noise or excessive vibration.	V	Personal injury/illness or hearing loss.
15. Using a hazardous chemical – chemicals, heavy metals, asbestos, Lead.	V	- Exposure to hazardous chemicals. - Spills / leaks (environmental).

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Key Steps in Logical Order		Hazards/Risks to Workers or Public	Risk Score	Control Measures	Residual Risk Score
1	Remove the current small blower and cyclone.	- The blower or cyclone could detach from rigging and fall, - Pinch or crush injuries - Falling debris -Electrical work.	High	- Inspect rigging equipment - Maintain safe distances from load or crane - Establish an exclusion zone beneath work area. - All electrical work to be done by licenced electricians.	Mod
2	Move large blower to work area with forklift.	-The blower may slip off the forks.	Low	- Use hardwood pallet - Avoid sudden movements	Low
2	Use crane to position large blower in place behind shredder.	- The blower may swing or shift unexpectedly - Workers in the tight space behind shredder could get caught between blower and shredder foundation. - The crane or load may strike the shedder leading to damage or injury.	Very High	- Minimize crane movements or operate at low speeds - Maintain clear communication between crane operator and ground workers - Use spotter to guide the crane in tight areas.	Mod
4	Use crane to bring cyclone from graveyard.	- Crane instability - Proximity to bucket elevators - Adverse weather conditions such as wind or rain may affect crane stability - Flora & fauna	High	- Conduct a ground survey to identify and weak or unstable surfaces - Plan the lift path to avoid obstacles. - Continuous monitoring of weather conditions - Check under and around cyclone for snakes or wasps etc.	Mod
5	Use crane to position and install new frame and cyclone on drag chain.	Crane instability The frame or cyclone could detach from rigging and fall. Workers could get fingers or hands crushed when attaching cyclone to drag chain	Very high	- Position the crane on level, stable ground and deploy outriggers - Workers could get fingers or hands crushed when attaching cyclone to drag chain. - Work to be carried out by competent workers gloves provided.	Mod
6	Use EWP to connect ducting from shredder to large New York blower and cyclone.	- Workers may fall from EWP basket while connecting the ducting - Tools or ducting components may fall from EWP causing injury to workers below - Workers may strain or injure themselves with positioning ducting.	Very High	- Wear fall arrest harness - Establish an exclusion zone below - Use team lifting techniques when handling ducting.	Mod

Attach additional pages if required

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All relevant workers to be consulted during this process and made aware of the control measures and all required permits identified.

AUTHORISATION			
Approved by	David Knight	Position	Plant manager
Signed		Date	

Name, Sign Off & Employer of Persons Performing Work		Start Time:	am/pm	Finish Time:	am/pm
Name	Sign Off	Employer			
1					
2					
3					
4					
5					

JOB COMPLETION					
1. Has equipment been modified/changed to affect how it operates?			☐ N (go to 3) ☐ Y (sign off by competent operator required)		
Has a <u>competent operator</u> approved it is safe to use? ☐ Y ☐ N		Competent Operator Name:		Competent Operator Signature:	
2. Work completed & area assessed safe before return to normal service? ☐ Y ☐ N <i>To be completed by the person conducting the task</i>			Name:		
			Signature:		
3. Altus Representative advised of & accepts changes: ☐ Y ☐ N		Name:		Signature:	

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